

Stochastic Gradient Descent for Polyak-Lojasiewicz Condition Functions

1 Algorithm Description

The algorithm iteratively updates parameters by computing gradients on randomly sampled data points and taking steps in the negative gradient direction with a fixed step size. It is designed for smooth functions satisfying the Polyak-Lojasiewicz condition.

1.1 Mathematical Formulation

The algorithm is designed to minimize a smooth function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfies the Polyak-Lojasiewicz (PL) condition. The PL condition states that there exists a constant $\mu > 0$ such that for all w ,

$$\frac{1}{2} \|\nabla f(w)\|^2 \geq \mu(f(w) - f^*),$$

where f^* is the minimum value of f .

Update Rule:

The algorithm updates the parameter vector w using the following rule:

1. Randomly sample a data point (or compute the gradient for a single deterministic function).
2. Compute the gradient $\nabla f(w)$.
3. Update the parameter:

$$w_{k+1} = w_k - \eta \nabla f(w_k),$$

where:

- w_k is the parameter vector at iteration k ,
- $\eta > 0$ is the fixed step size (learning rate),
- $\nabla f(w_k)$ is the gradient of f at w_k .

Hyperparameters:

- **Step size (η):** Determines the size of each update step. It should be chosen to satisfy convergence criteria, typically small enough to ensure stability.
- **Minibatch size:** Number of data points used to compute the gradient. Here it is set to 1 for simplicity (fully stochastic).

2 Pseudo Code and Dry Run

2.1 Pseudocode

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Initialize: w = w0, k = 0
while not converged do
    Sample a data point (or use deterministic function)
    Compute gradient: grad = \nabla f(w)
    Update parameters: w = w - \eta * grad
    k = k + 1
    Check convergence criteria (e.g., norm of gradient below threshold)
end while

```

2.2 Dry Run Example

Consider the function $f(w) = (w - 3)^2$.

Initialization:

- $w_0 = 0$
- Step size $\eta = 0.1$
- Minibatch size = 1

Iteration 1:

- Compute gradient: $\nabla f(w_0) = 2(w_0 - 3) = 2(0 - 3) = -6$
- Update parameter: $w_1 = w_0 - \eta \nabla f(w_0) = 0 - 0.1 \times (-6) = 0.6$
- Compute objective value: $f(w_1) = (0.6 - 3)^2 = 5.76$

Iteration 2:

- Compute gradient: $\nabla f(w_1) = 2(w_1 - 3) = 2(0.6 - 3) = -4.8$
- Update parameter: $w_2 = w_1 - \eta \nabla f(w_1) = 0.6 - 0.1 \times (-4.8) = 1.08$
- Compute objective value: $f(w_2) = (1.08 - 3)^2 = 3.6864$

Iteration 3:

- Compute gradient: $\nabla f(w_2) = 2(w_2 - 3) = 2(1.08 - 3) = -3.84$

- Update parameter: $w_3 = w_2 - \eta \nabla f(w_2) = 1.08 - 0.1 \times (-3.84) = 1.464$
- Compute objective value: $f(w_3) = (1.464 - 3)^2 = 2.3593$

By following these steps, the algorithm iteratively moves w closer to the optimal value of 3, minimizing $f(w)$.

3 Convergence Theory

3.1 Assumptions

We consider the optimization problem of minimizing a smooth function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfies the Polyak-Lojasiewicz (PL) condition. The following assumptions are made:

- **Smoothness:** The function f is L -smooth, i.e., for all $x, y \in \mathbb{R}^n$, we have:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|.$$

- **Polyak-Lojasiewicz Condition:** There exists a constant $\mu > 0$ such that for all $x \in \mathbb{R}^n$, the following holds:

$$\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu(f(x) - f^*),$$

where f^* is the global minimum of f .

- **Fixed Step Size:** The algorithm uses a constant step size $\alpha > 0$.
- **Random Sampling:** Gradients are computed on randomly sampled data points, following the assumption of unbiased stochastic gradient:

$$\mathbb{E}[\nabla \tilde{f}(x)] = \nabla f(x),$$

where $\nabla \tilde{f}(x)$ is the stochastic gradient.

3.2 Main Convergence Theorem

Theorem: Under the assumptions stated above, let $\{x_k\}$ be the sequence generated by the algorithm, with $x_{k+1} = x_k - \alpha \nabla \tilde{f}(x_k)$. If the step size α satisfies $0 < \alpha < \frac{2}{L}$, then $\{x_k\}$ converges to x^* with a linear convergence rate. Specifically, we have:

$$\mathbb{E}[f(x_k) - f^*] \leq (1 - 2\alpha\mu)^k (f(x_0) - f^*).$$

3.3 Theoretical Properties

3.3.1 Stability Analysis

The stability of the algorithm is influenced by the choice of the step size α . A smaller α improves stability but may slow convergence, while a larger α risks divergence if $\alpha \geq \frac{2}{L}$.

3.3.2 Sensitivity to Hyperparameters

The convergence rate depends on the condition number $\kappa = \frac{L}{\mu}$. A better conditioned problem (smaller κ) leads to faster convergence. The step size α must be chosen carefully within the range $(0, \frac{2}{L})$.

3.3.3 Comparison to Baseline Methods

Compared to methods that do not assume the PL condition, this algorithm achieves faster convergence under the same smoothness assumption. While methods like standard gradient descent achieve $O(1/\epsilon)$ convergence in terms of iteration complexity, our method achieves linear convergence.

3.4 Discussion

The convergence theory developed here demonstrates that the algorithm efficiently minimizes smooth functions under the PL condition with a linear rate, given an appropriate choice of step size. This highlights the importance of leveraging problem structure to enhance optimization performance. Future work may involve extending this framework to non-convex settings where the PL condition holds locally.

4 Convergence Proof

4.1 Preliminaries (Definitions and Lemmas)

4.1.1 Definitions

1. **Smoothness:** A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth if for all $x, y \in \mathbb{R}^n$:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|.$$

2. **Polyak-Lojasiewicz Condition:** A function f satisfies the PL condition if there exists $\mu > 0$ such that for all $x \in \mathbb{R}^n$:

$$\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu(f(x) - f^*),$$

where f^* is the global minimum value of f .

3. **Stochastic Gradient:** The stochastic gradient $\nabla \tilde{f}(x)$ is an unbiased estimator of the true gradient:

$$\mathbb{E}[\nabla \tilde{f}(x)] = \nabla f(x).$$

4.1.2 Lemmas

- **Lemma 1 (Descent Lemma):** For an L -smooth function f , the following holds for any $x, y \in \mathbb{R}^n$:

$$f(y) \leq f(x) + \nabla f(x)^T(y - x) + \frac{L}{2}\|y - x\|^2.$$

- **Lemma 2 (PL Condition Implication):** If f satisfies the PL condition, then:

$$\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f^*).$$

4.2 Main Proof (Step-by-Step Derivation)

We aim to prove that the sequence $\{x_k\}$ generated by the update rule $x_{k+1} = x_k - \alpha \nabla \tilde{f}(x_k)$ converges linearly to x^* .

1. **Expected Descent:**

$$\begin{aligned} \mathbb{E}[f(x_{k+1})] &\leq \mathbb{E}\left[f(x_k) + \nabla f(x_k)^T(x_{k+1} - x_k) + \frac{L}{2}\|x_{k+1} - x_k\|^2\right] \\ &= \mathbb{E}\left[f(x_k) - \alpha \nabla f(x_k)^T \nabla \tilde{f}(x_k) + \frac{L}{2}\alpha^2 \|\nabla \tilde{f}(x_k)\|^2\right]. \end{aligned}$$

2. **Using Unbiasedness:**

$$\mathbb{E}[\nabla f(x_k)^T \nabla \tilde{f}(x_k)] = \|\nabla f(x_k)\|^2.$$

3. **Substituting and Simplifying:**

$$\mathbb{E}[f(x_{k+1})] \leq f(x_k) - \alpha \|\nabla f(x_k)\|^2 + \frac{L}{2}\alpha^2 \mathbb{E}[\|\nabla \tilde{f}(x_k)\|^2].$$

4. **Bounding the Variance:** Assume $\mathbb{E}[\|\nabla \tilde{f}(x_k)\|^2] \leq \|\nabla f(x_k)\|^2 + \sigma^2$.

5. **Combining Terms:**

$$\begin{aligned} \mathbb{E}[f(x_{k+1})] &\leq f(x_k) - \alpha \|\nabla f(x_k)\|^2 + \frac{L}{2}\alpha^2 (\|\nabla f(x_k)\|^2 + \sigma^2) \\ &= f(x_k) - \left(\alpha - \frac{L}{2}\alpha^2\right) \|\nabla f(x_k)\|^2 + \frac{L}{2}\alpha^2 \sigma^2. \end{aligned}$$

6. **Applying PL Condition:**

$$\|\nabla f(x_k)\|^2 \geq 2\mu(f(x_k) - f^*).$$

7. **Substituting PL Condition:**

$$\mathbb{E}[f(x_{k+1}) - f^*] \leq (1 - 2\alpha\mu + L\alpha^2\mu)(f(x_k) - f^*) + \frac{L}{2}\alpha^2\sigma^2.$$

8. **Choosing Step Size:** For linear convergence, choose α such that $0 < \alpha < \frac{2}{L}$.

4.3 Rate Derivation (With Constants)

1. **Linear Convergence Rate:**

$$\mathbb{E}[f(x_k) - f^*] \leq (1 - 2\alpha\mu + L\alpha^2\mu)^k (f(x_0) - f^*) + \frac{L}{2}\alpha^2\sigma^2 \sum_{i=0}^{k-1} (1 - 2\alpha\mu + L\alpha^2\mu)^i.$$

2. **Simplifying the Geometric Series:**

$$\sum_{i=0}^{k-1} (1 - 2\alpha\mu + L\alpha^2\mu)^i \leq \frac{1}{2\alpha\mu - L\alpha^2\mu}.$$

3. **Final Rate:**

$$\mathbb{E}[f(x_k) - f^*] \leq (1 - 2\alpha\mu + L\alpha^2\mu)^k (f(x_0) - f^*) + \frac{L\alpha^2\sigma^2}{2(2\alpha\mu - L\alpha^2\mu)}.$$

4.4 Special Cases

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Gradient Descent **Deterministic Gradient**

Gradient Descent: If $\sigma^2 = 0$, the convergence rate simplifies to:

$$\mathbb{E}[f(x_k) - f^*] \leq (1 - 2\alpha\mu + L\alpha^2\mu)^k (f(x_0) - f^*).$$

- **Effect of Condition Number:** The convergence rate depends on the condition number $\kappa = \frac{L}{\mu}$. A smaller κ results in faster convergence.

This proof demonstrates that under the PL condition and with an appropriate choice of step size, the algorithm achieves linear convergence. The detailed derivation provides insights into the role of each assumption and parameter in the convergence behavior.